
H1N1 Vaccination Analysis by Valary Kones

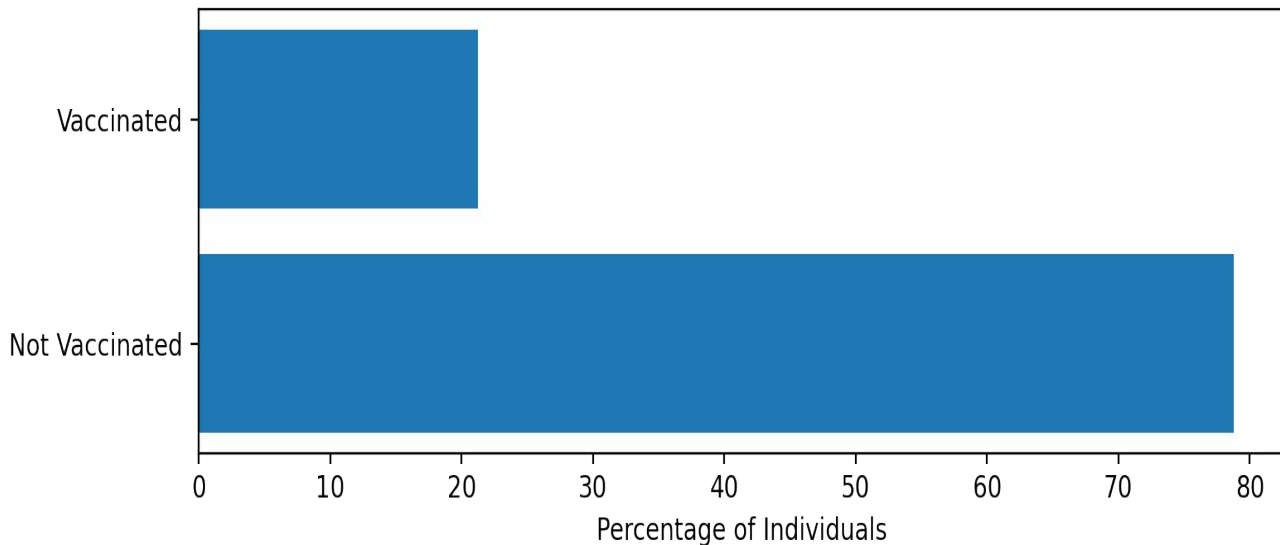
Identifying Key Drivers to Boost Vaccination Rates

Business Problem

- H1N1 vaccination rates remain low
- Public health campaigns rely on **broad messaging**
- Limited Resources — outreach must be **targeted and effective**

Data Understanding

H1N1 Vaccination Class Distribution



Data Source

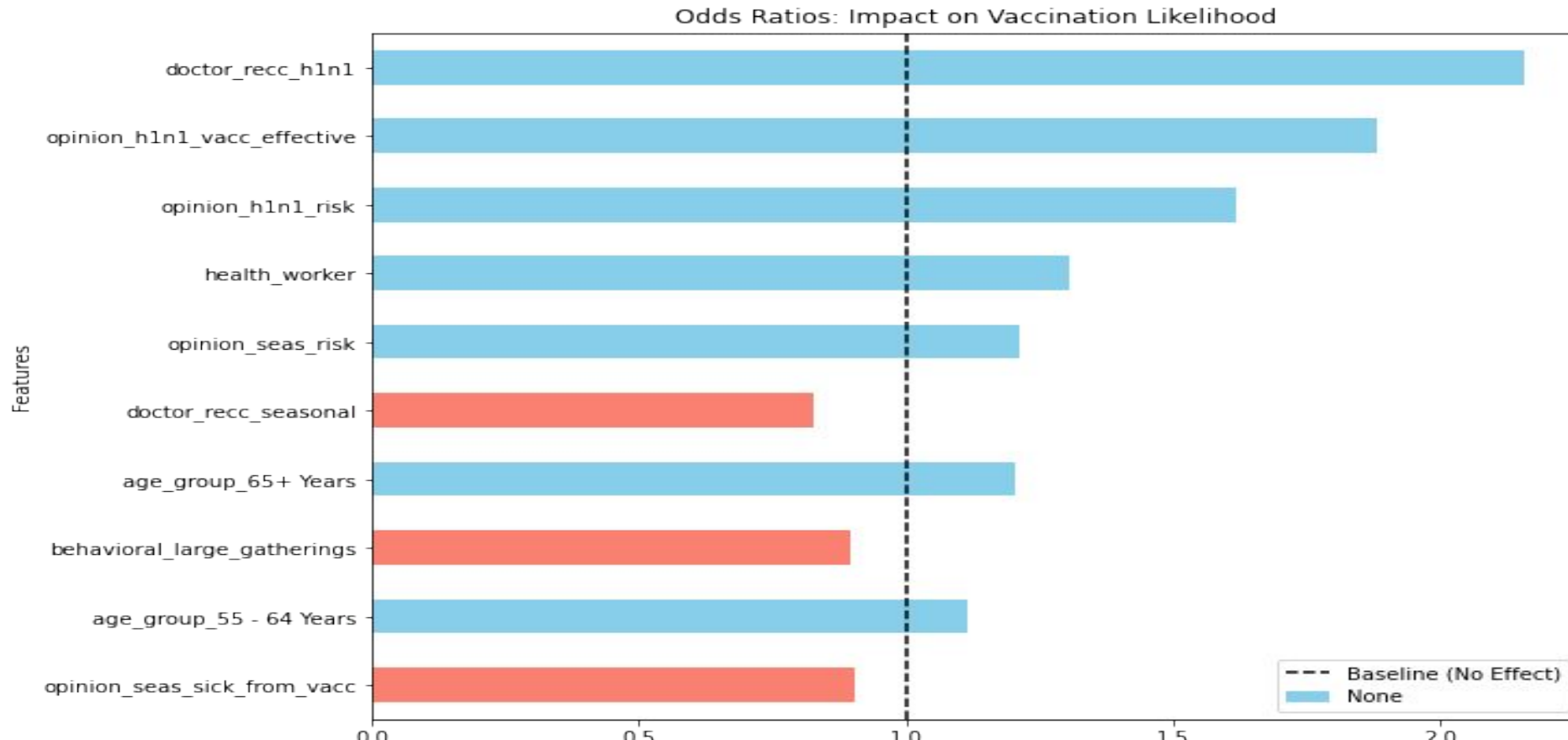
- *National 2009 H1N1 Flu Survey*

Important features:

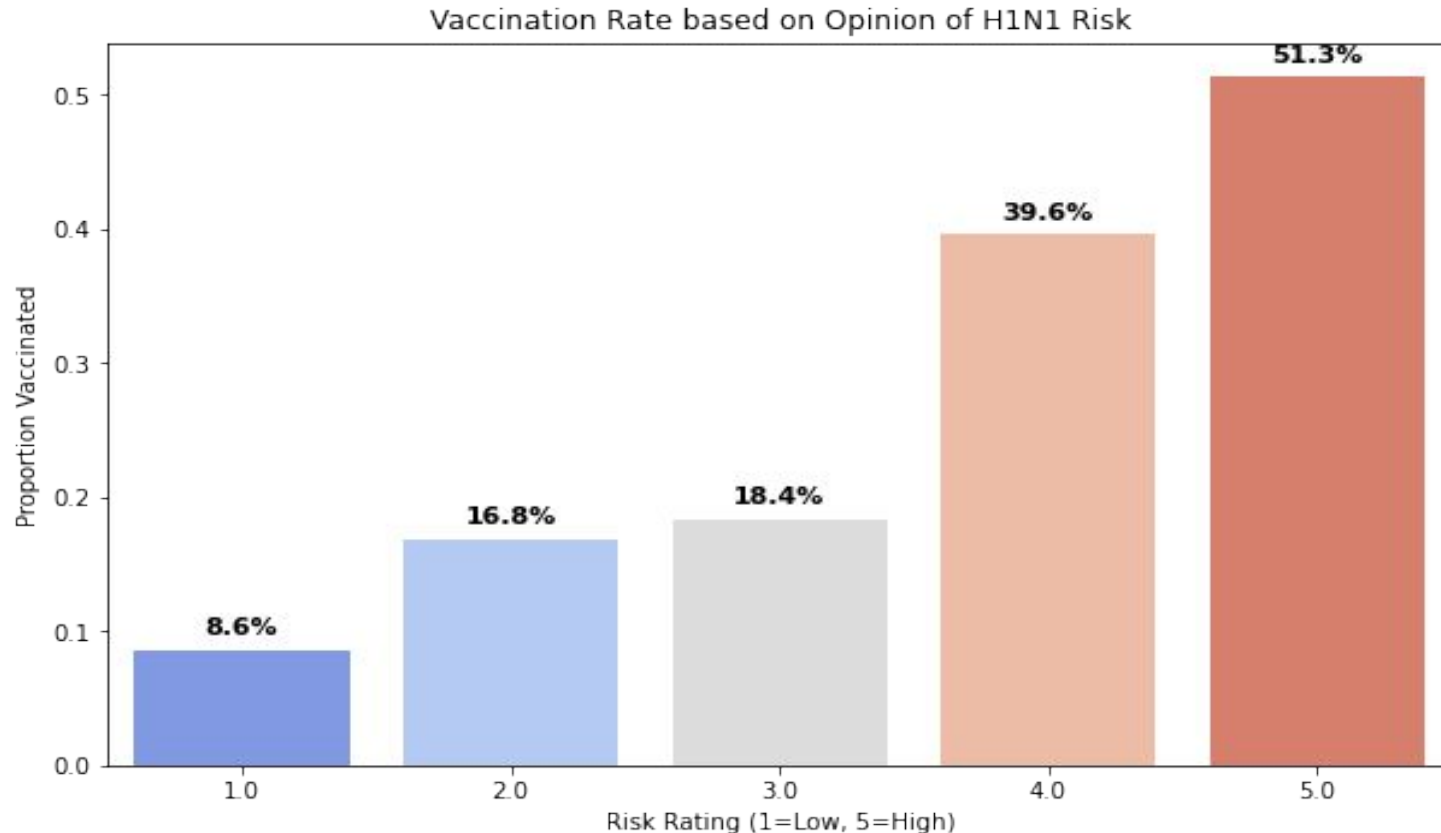
- *Doctor's rec*
- *Risk Perception*
- *Personal Beliefs*

Exploratory Data Analysis

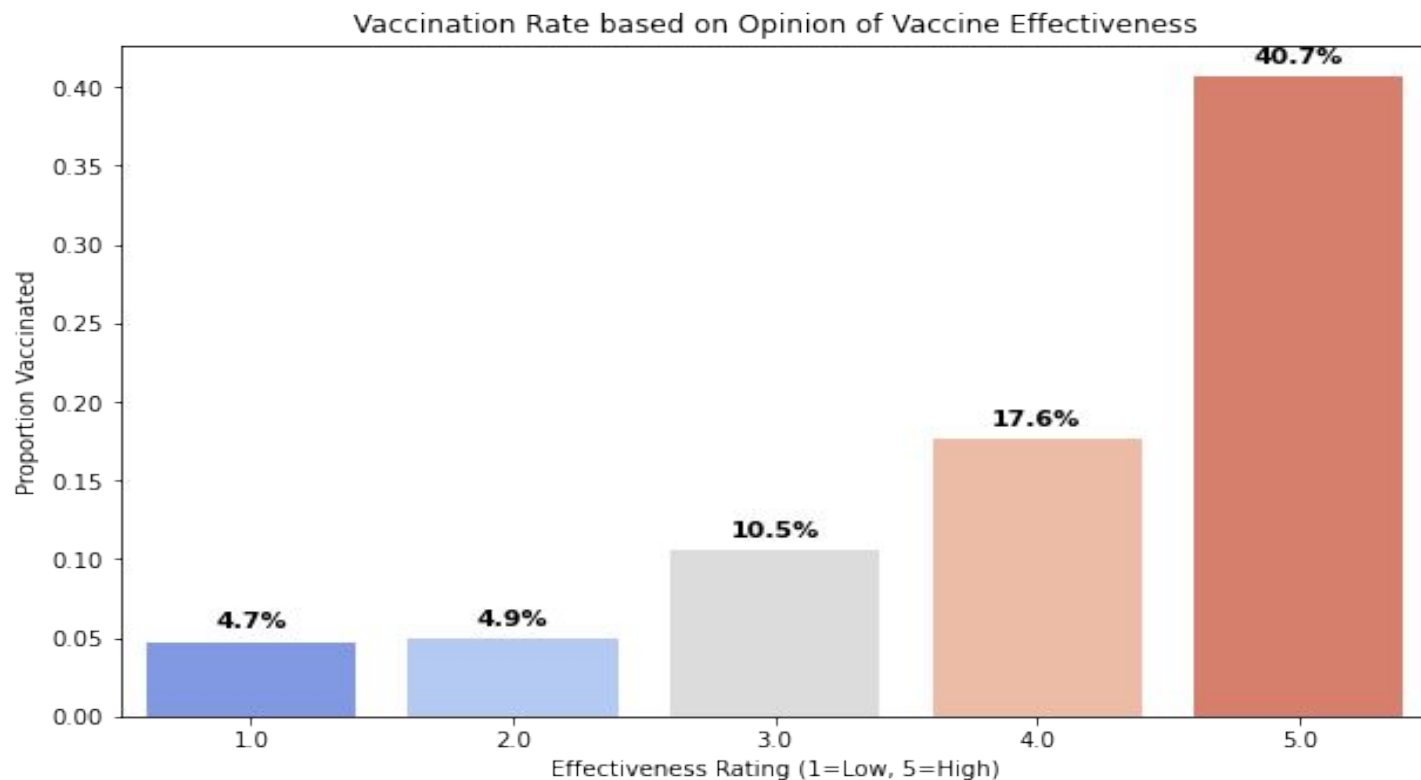
Doctor's Recommendation: The Biggest Driver



Risk Perception Shapes Behaviour

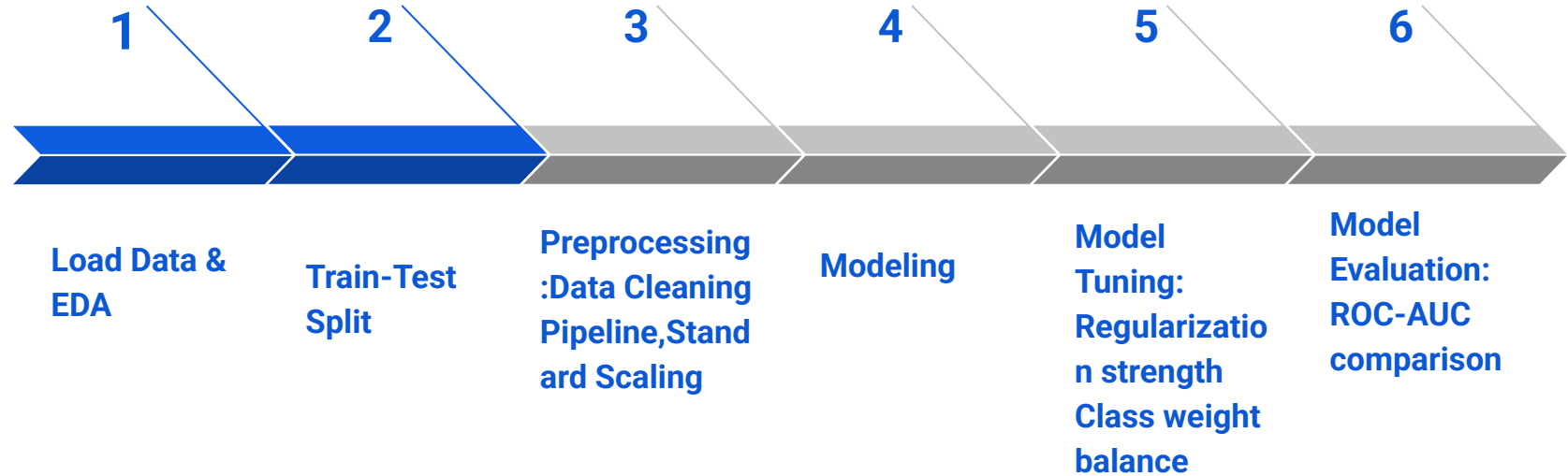


Trust In Vaccine Effectiveness



Modeling

Modeling Process



Model Performance

Model Performance Comparison

Model	Train AUC	Test AUC	CV Mean AUC	Overfit Gap
Baseline (Logistic Regression)	0.8346	0.8242	0.8317	0.0104
Tuned Logistic Regression ☐	0.8337	0.8240	0.8313	0.0097
Vanilla Decision Tree	0.8236	0.8070	0.8168	0.0166
Tuned Decision Tree	0.8230	0.8095	0.8169	0.0134

☐ Best overall model based on ROC-AUC and generalization performance

Recommendations

- **Empower doctors** to recommend vaccines consistently
- **Address safety fears directly**, not just promote benefits
- **Target low risk perception**, not demographics

Future Steps

- **Improve targeting strategies**
- **Use risk scoring for outreach prioritization**
- **Apply approach to future vaccination campaigns**

Thank You!
